

Update Joint Modeling 18/08

Overview Methods:

Three methods for estimating model parameters were suggested:

1. Classical maximum likelihood
2. Multiple imputation + closed form
3. Closed form + weighting

Currently:

1. Classical ML completely implemented
2. MI + closed form partly implemented: Fixed effects and variance is implemented, RE and variance estimators of fixed effects **implemented but check D-matrix**
3. Closed form + weighting: **not yet implemented Q: iterative reweighting? How many iterations? What convergence criterion**

Preliminary Results

Simulation setup

Formula:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 Time_j + \beta_3 Treatment_i * Time_j + b_{0i} + b_{1i} * Time_j$$

Parameters were taken from the toenail data:

Effect	Parameter	Size
Intercept	β_0	2.4562
Treatment Effect	β_1	0 (0.593 in toenail data (this was a dose effect))

Effect	Parameter	Size
Time Effect	β_2	0.2792
Treatment x Time	β_3	0.0350
Random intercept variance	d_{11}	7.3174
Random slope variance	d_{22}	0.2239
Random effects covariance	d_{12}	-0.4985
Residual variance	σ^2	3.1508

Variable	Value
number of clusters	100, 50, 20, 10 (<i>10.000 clusters was asked too, but currently excluded due to large computing time</i>)
number of repeated measures	12
missingness pattern	“half-missing” (half of the clusters have no missingness, other half have uniform probability to have 1-11 observations, dropout is monotone)
number of simulations / scenario	100 (<i>for illustration, will use 1000 or more in the future, but restricted due to computing time</i>)

For the imputations: imputations were done within clusters, 3 imputations per clusters, with a maximum of 10 repetitions.

Next steps

1. Check rate of convergence
2. Implement Type I error rate
3. Implement closed form + weighting
4. Present results with graphs
5. Joint model

Results

Mean Squared Error

	method	n	mse_beta0	mse_beta1	mse_beta2	mse_beta3
1	classical_ml	100	0.1815752	0.3076049	0.008084674	0.01485622
5	mi_closed_form	100	0.1831959	0.3055200	0.006745952	0.01017153
2	classical_ml	50	0.3139378	0.6059362	0.011491919	0.02500252
6	mi_closed_form	50	0.3055302	0.6033598	0.008791732	0.01704832
3	classical_ml	20	0.9832979	1.8367591	0.033316005	0.07408942
7	mi_closed_form	20	0.9533063	1.6765302	0.028370005	0.04454796
4	classical_ml	10	1.9374200	4.2105567	0.077038354	0.13120062
8	mi_closed_form	10	1.8976943	4.0957131	0.058333856	0.08263798

Mean Absolute Bias

	method	n	mean_abs_bias_beta0	mean_abs_bias_beta1
1	classical_ml	100	0.3462758	0.4418633
5	mi_closed_form	100	0.3452351	0.4372822
2	classical_ml	50	0.4553941	0.6075810
6	mi_closed_form	50	0.4444811	0.6108699
3	classical_ml	20	0.7635551	1.0419777
7	mi_closed_form	20	0.7481704	1.0015089
4	classical_ml	10	1.0675673	1.5962527
8	mi_closed_form	10	1.0680856	1.5929839
	mean_abs_bias_beta2		mean_abs_bias_beta3	
1	0.07012774		0.09621034	
5	0.06541233		0.07991887	
2	0.08541736		0.12894778	
6	0.07433543		0.10517760	
3	0.14930150		0.21609614	
7	0.13359287		0.16757761	
4	0.23012221		0.28725325	
8	0.19924579		0.22258852	

Mean Relative Bias

Note: probably better to use mean absolute relative bias?

	method	n	mean_rel_bias_beta0	mean_rel_bias_beta1
1	classical_ml	100	-0.016570988	NA

5	mi_closed_form	100	-0.017764785	NA
2	classical_ml	50	-0.022272205	NA
6	mi_closed_form	50	-0.022915646	NA
3	classical_ml	20	0.001812355	NA
7	mi_closed_form	20	0.001713899	NA
4	classical_ml	10	0.021033969	NA
8	mi_closed_form	10	0.037380498	NA
	mean_rel_bias_beta2		mean_rel_bias_beta3	
1		0.02024501	-0.3723699	
5		0.02422089	-0.5222395	
2		0.03437231	-0.8421612	
6		0.02753804	-0.7977347	
3		0.10953193	-0.7885311	
7		0.11388355	-0.6475062	
4		0.21931469	-2.2492589	
8		0.17787113	-1.6644295	

Convergence Rate

Note: check convergence rate for imputation

	method	n	prop_converged_ok	prop_converged_warning
1	classical_ml	100	0.96	0.04
5	mi_closed_form	100	0.00	0.00
2	classical_ml	50	0.95	0.05
6	mi_closed_form	50	0.00	0.00
3	classical_ml	20	1.00	0.00
7	mi_closed_form	20	0.00	0.00
4	classical_ml	10	0.89	0.01
8	mi_closed_form	10	0.00	0.00
	prop_converged_singular			
1			0	
5			0	
2			0	
6			0	
3			0	
7			0	
4			0	
8			0	

Coverage rate of β_3

	method	n	coverage95_beta3
1	classical_ml	100	0.96
5	mi_closed_form	100	0.94
2	classical_ml	50	0.96
6	mi_closed_form	50	0.92
3	classical_ml	20	0.91
7	mi_closed_form	20	0.92
4	classical_ml	10	0.85
8	mi_closed_form	10	0.94

Computation Time

Classical ML is faster (due to time required for imputation)

	method	n	time_mean_seconds	time_median_seconds
1	classical_ml	100	0.0279	0.030
5	mi_closed_form	100	1.4016	1.340
2	classical_ml	50	0.0208	0.020
6	mi_closed_form	50	1.1208	0.990
3	classical_ml	20	0.0171	0.020
7	mi_closed_form	20	0.8432	0.800
4	classical_ml	10	0.0171	0.020
8	mi_closed_form	10	0.8641	0.815